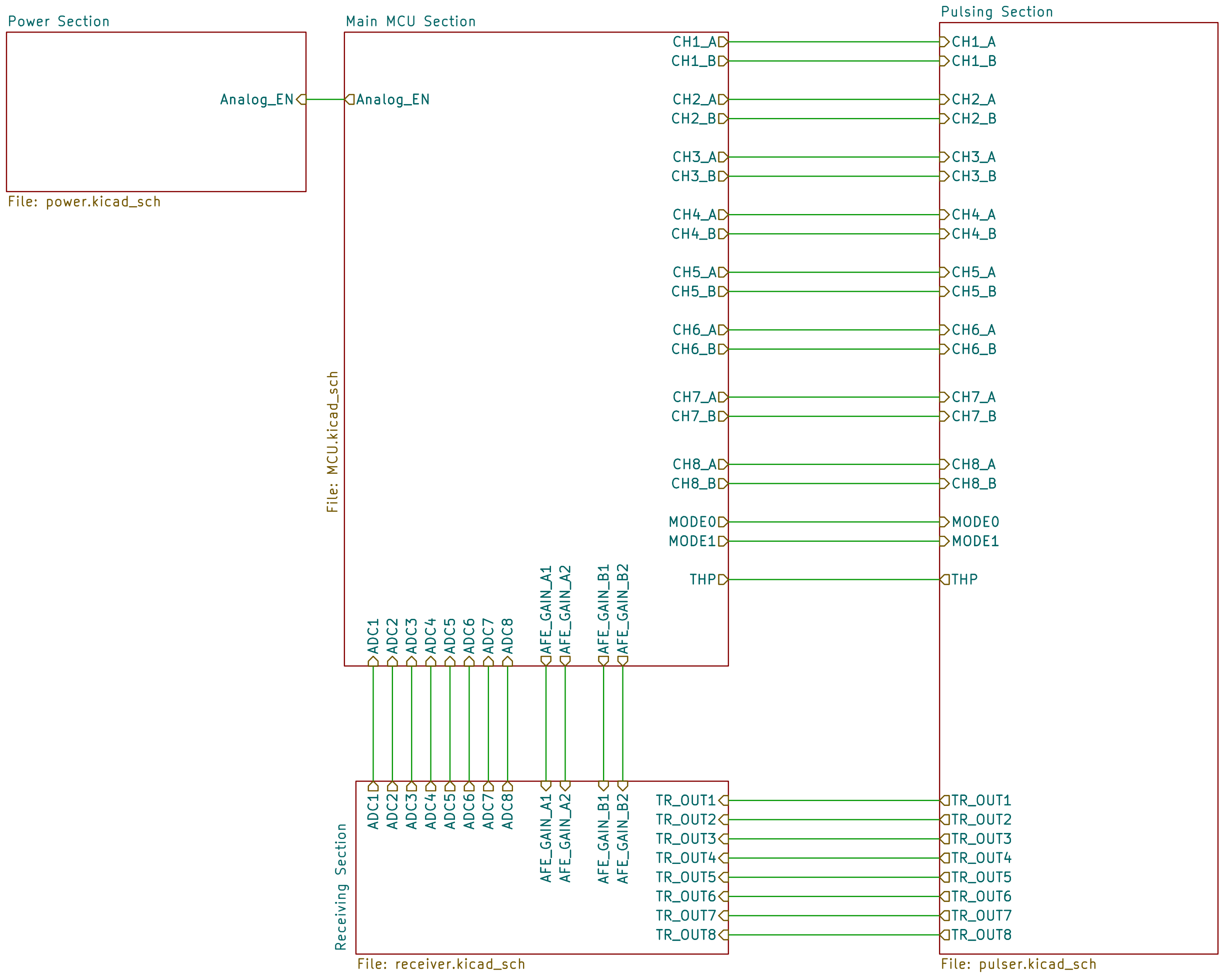



| | | | |
|---|---|---|---|
| **12 Volt** (External) (**555mA**) | **7 Volt** (Buck Converter) (**350mA**) | **5 Volt** (LDO Regulator) (**200mA**) | LNA + VGA Supply (**160mA** – short circuit) |
| | | | Op-Amps (4 * 10mA = **40mA**) |
| | | **3.3 Volt** (LDO Regulator) (**150mA**) | MCU Supply (**150mA**) |
| | **5 Volt** (Buck Converter) (**205mA**) | Positive Pulse Driver Supply (**100mA**) | |
| | | **3.3 Volt** (LDO Regulator) (**5mA**) | Pulser Logic Supply (**5mA**) |
| | | **-5 Volt** (Inverting Buck Converter) (**100mA**) | Negative Pulse Driver Supply (**100mA**) |

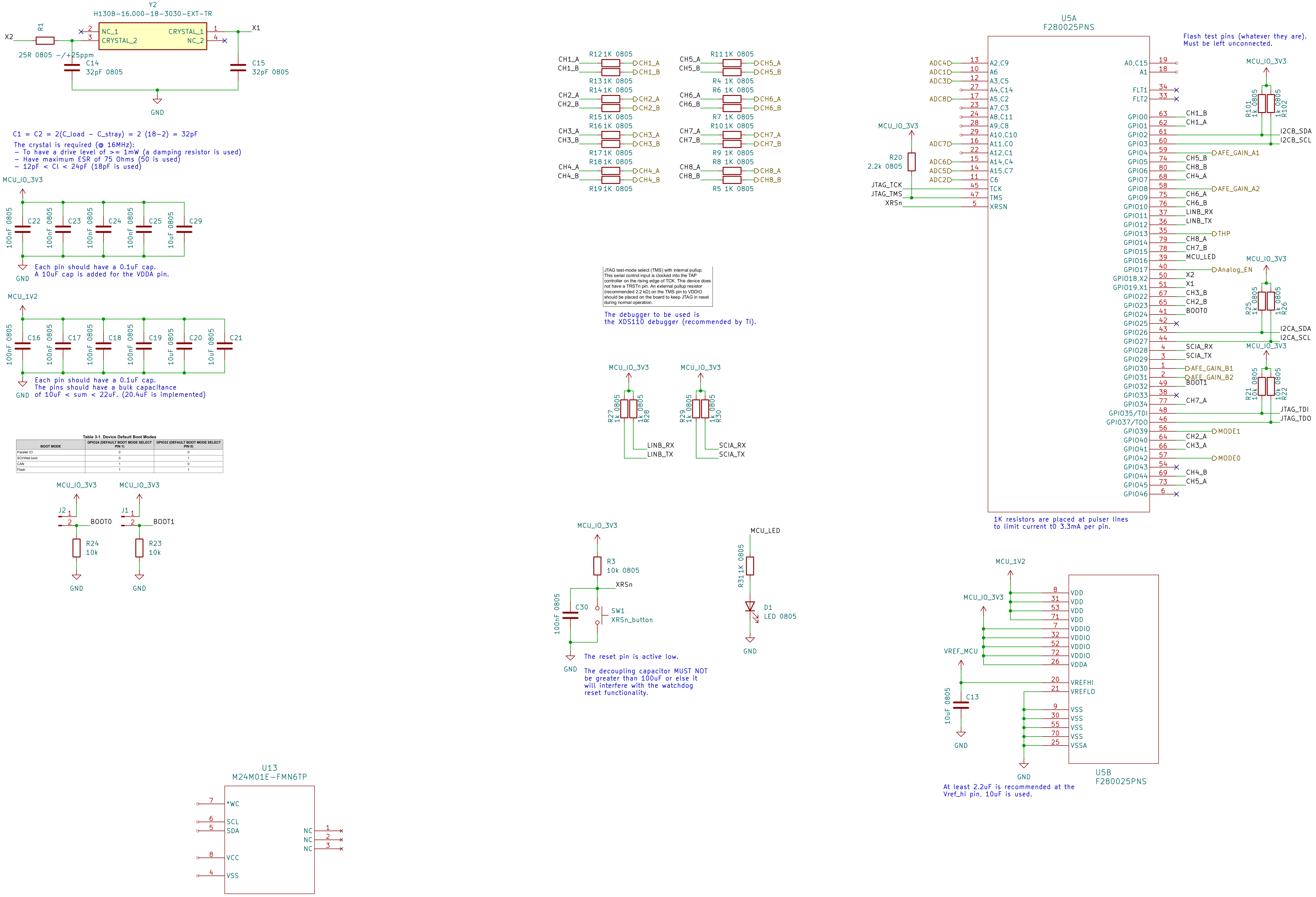
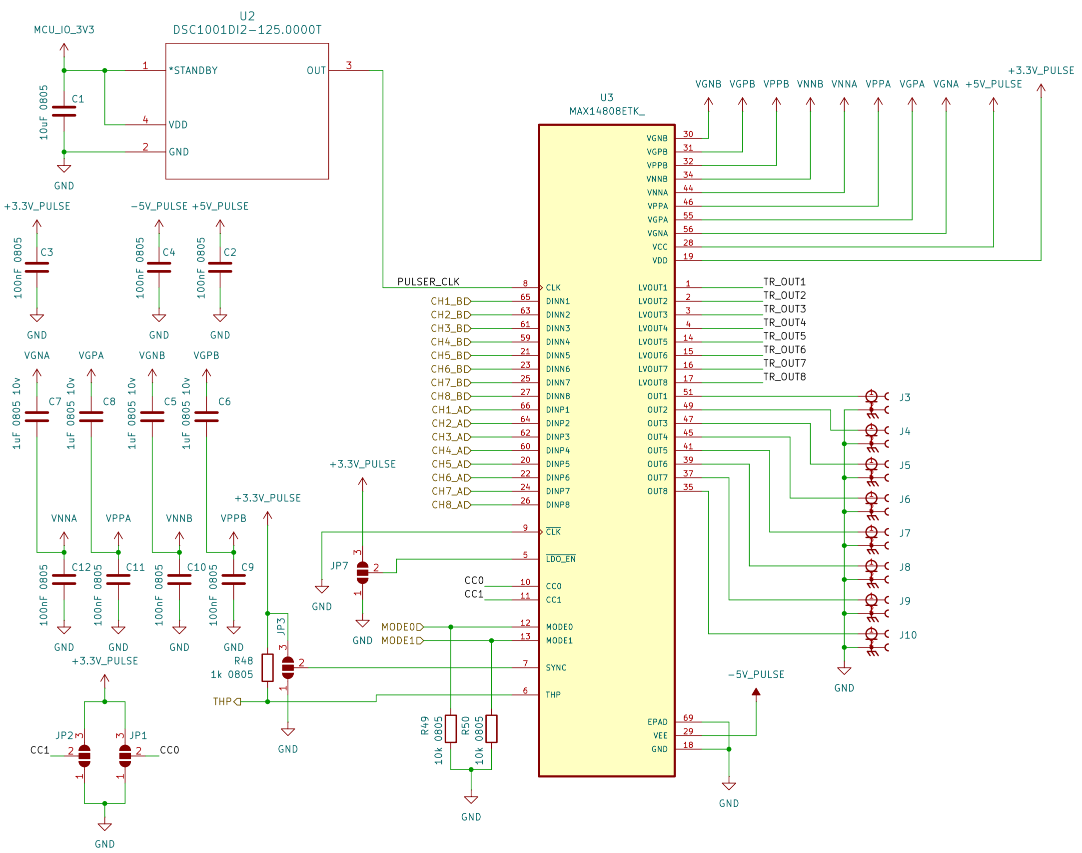
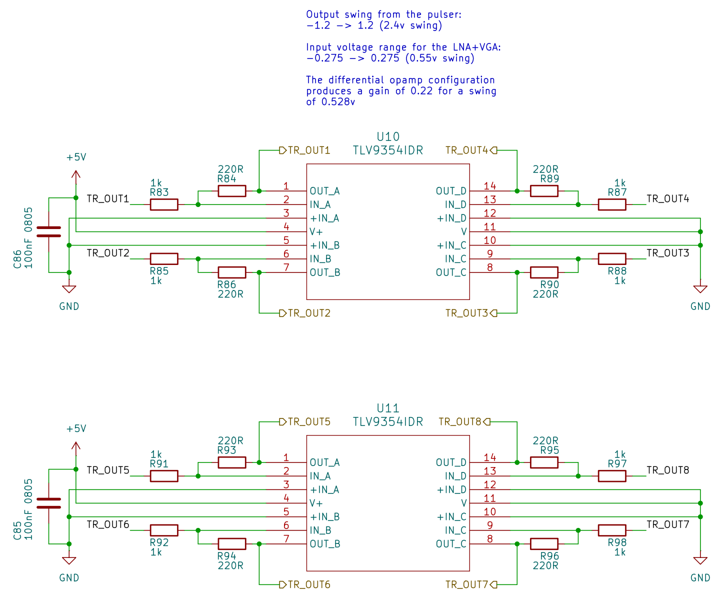



Table 3-1. Device Default Boot Modes

| BOOT MODE | GPIO24 (DEFAULT BOOT MODE SELECT PIN 1) | GPIO32 (DEFAULT BOOT MODE SELECT PIN 0) |
|---|---|---|
| Parallel IO | 0 | 0 |
| SCI/Wait boot | 0 | 1 |
| CAN | 1 | 0 |
| Flash | 1 | 1 |




VGXX: gate driver voltage supplies
(positive and negative for ports A & B)

VNNX/VPPX: positive and negative HV supplies
(for ports A & B)

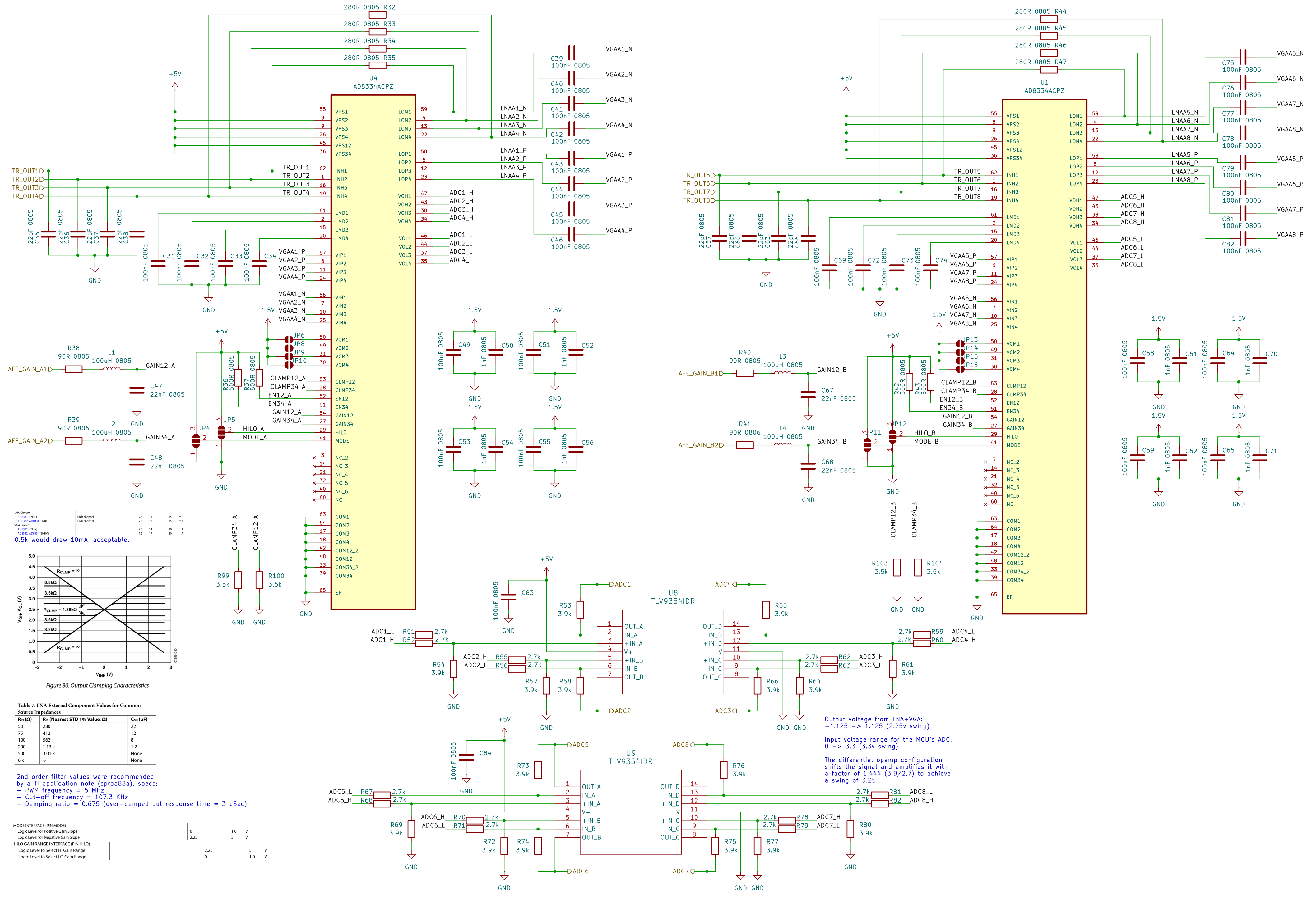
**Enabling/Disabling the internal floating power supplies (FPS):**

requirements in most of the cases. Drive $\overline{\text{LDO\_EN}}$ low or leave unconnected to enable the internal FPSs; drive $\overline{\text{LDO\_EN}}$ high to disable the internal FPSs.

A triple solder jumper is placed
to either pull-up (select external FPS)
or pull-down (select internal FPS) the
LDOEN pin.

**Table 5. Current Drive Selection**

| INPUTS | | PULSER OUTPUT CURRENT (typ) |
|---|---|---|
| CC0 | CC1 | |
| 0 | 0 | 2A |
| 1 | 0 | 1.5A |
| 0 | 1 | 1A |
| 1 | 1 | 0.5A |




| | | | | | |
|---|---|---|---|---|---|
| LNA Current | | | | | |
| AD8331 (ENBL) | Each channel | 7.5 | 11 | 15 | mA |
| AD8332, AD8334 (ENBL) | Each channel | 7.5 | 12 | 15 | mA |
| VGA Current | | | | | |
| AD8331 (ENBV) | | 7.5 | 14 | 20 | mA |
| AD8332, AD8334 (ENBV) | | 7.5 | 17 | 20 | mA |

0.5k would draw 10mA, acceptable.

*Figure 80. Output Clamping Characteristics*

**Table 7. LNA External Component Values for Common Source Impedances**

| $R_{IN}$ (Ω) | $R_{IZ}$ (Nearest STD 1% Value, Ω) | $C_{SH}$ (pF) |
|---|---|---|
| 50 | 280 | 22 |
| 75 | 412 | 12 |
| 100 | 562 | 8 |
| 200 | 1.13 k | 1.2 |
| 500 | 3.01 k | None |
| 6 k | ∞ | None |

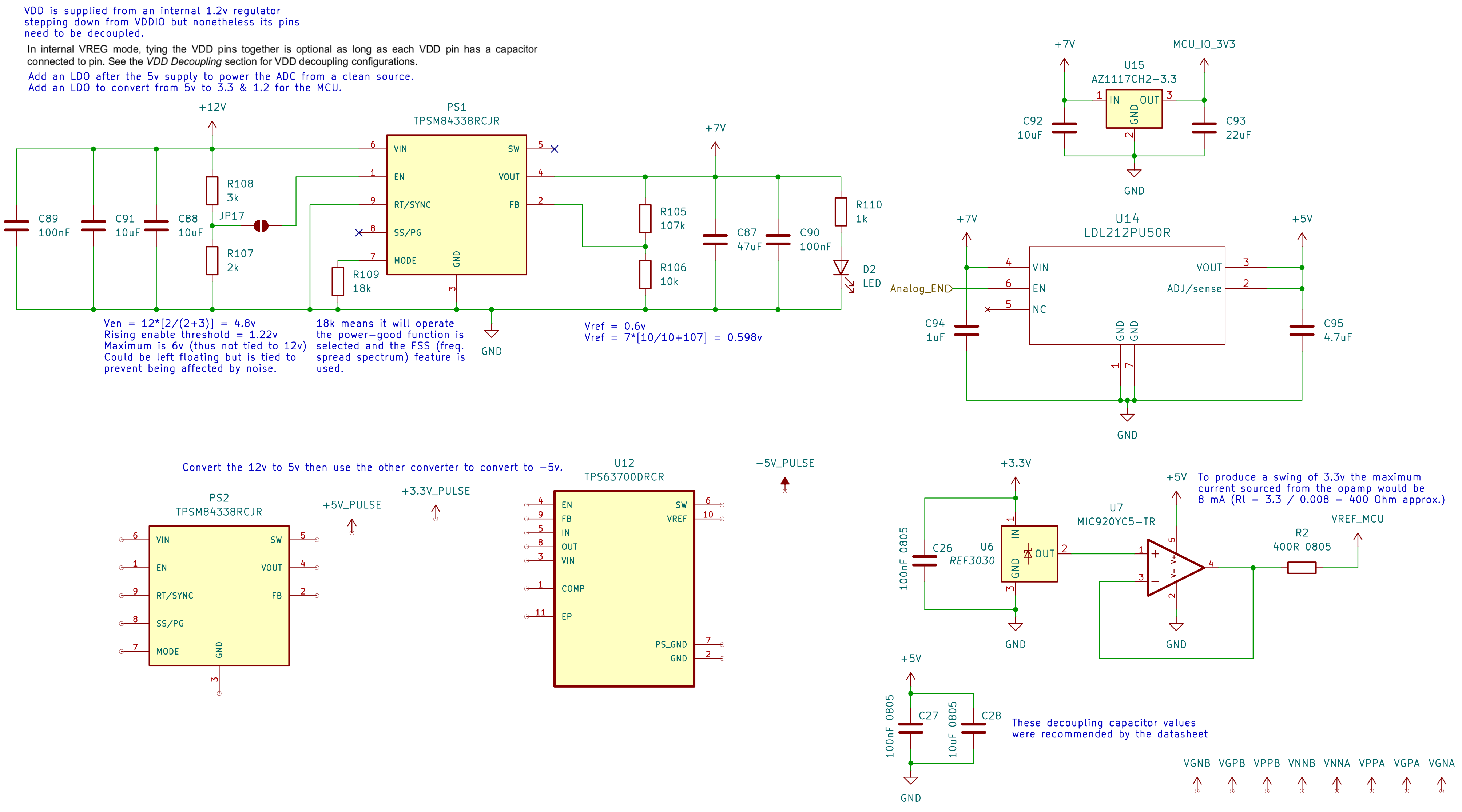
2nd order filter values were recommended
by a TI application note (spraa88a), specs:
- PWM frequency = 5 MHz
- Cut–off frequency = 107.3 KHz
- Damping ratio = 0.675 (over–damped but response time = 3 uSec)

| | | | |
|---|---|---|---|
| MODE INTERFACE (PIN MODE) | | | |
| Logic Level for Positive Gain Slope | 0 | 1.0 | V |
| Logic Level for Negative Gain Slope | 2.25 | 5 | V |
| HILO GAIN RANGE INTERFACE (PIN HILO) | | | |
| Logic Level to Select HI Gain Range | 2.25 | 5 | V |
| Logic Level to Select LO Gain Range | 0 | 1.0 | V |

Output voltage from LNA+VGA:
–1.125 –> 1.125 (2.25v swing)

Input voltage range for the MCU's ADC:
0 –> 3.3 (3.3v swing)

The differential opamp configuration shifts the signal and amplifies it with a factor of 1.444 (3.9/2.7) to achieve a swing of 3.25.


VDD is supplied from an internal 1.2v regulator
stepping down from VDDIO but nonetheless its pins
need to be decoupled.
In internal VREG mode, tying the VDD pins together is optional as long as each VDD pin has a capacitor
connected to pin. See the VDD Decoupling section for VDD decoupling configurations.
Add an LDO after the 5v supply to power the ADC from a clean source.
Add an LDO to convert from 5v to 3.3 & 1.2 for the MCU.
+12V
PS1
TPSM84338RCJR
VIN
EN
RT/SYNC
SS/PG
MODE
GND
SW
VOUT
FB
+7V
C89
100nF
C91
10uF
C88
10uF
R108
3k
JP17
R107
2k
R109
18k
R105
107k
R106
10k
C87
47uF
C90
100nF
R110
1k
D2
LED
GND
Ven = 12*[2/(2+3)] = 4.8v
Rising enable threshold = 1.22v
Maximum is 6v (thus not tied to 12v)
Could be left floating but is tied to
prevent being affected by noise.
18k means it will operate
the power-good function is
selected and the FSS (freq.
spread spectrum) feature is
used.
Vref = 0.6v
Vref = 7*[10/10+107] = 0.598v
+7V
MCU_IO_3V3
U15
AZ1117CH2-3.3
IN
OUT
GND
C92
10uF
C93
22uF
GND
+7V
U14
LDL212PU50R
+5V
VIN
EN
NC
VOUT
ADJ/sense
GND
GND
Analog_EN
C94
1uF
C95
4.7uF
GND
Convert the 12v to 5v then use the other converter to convert to -5v.
PS2
TPSM84338RCJR
VIN
EN
RT/SYNC
SS/PG
MODE
GND
SW
VOUT
FB
+5V_PULSE
+3.3V_PULSE
U12
TPS63700DRCR
EN
FB
IN
OUT
VIN
COMP
EP
SW
VREF
PS_GND
GND
-5V_PULSE
+3.3V
C26
100nF 0805
U6
REF3030
IN
OUT
GND
GND
+5V
U7
MIC920YC5-TR
GND
To produce a swing of 3.3v the maximum
current sourced from the opamp would be
8 mA (Rl = 3.3 / 0.008 = 400 Ohm approx.)
VREF_MCU
R2
400R 0805
+5V
C27
100nF 0805
C28
10uF 0805
These decoupling capacitor values
were recommended by the datasheet
GND
VGNB VGPB VPPB VNNB VNNA VPPA VGPA VGNA